ASYMPTOTIC NORM OF THE BLOCH-TO-BERGMAN INCLUSION

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Evidence is labelled Sketch: this proof has not been posted on arXiv or peer reviewed and should be checked by the reader. AI assistance was used in adapting and checking the argument and exposition.

> "Conjecture 3.9. For any $\alpha > -1$,
>
> $$\lim_{p\to\infty} \frac{C_\alpha(p)}{p/(2e)} = \frac{1}{\sqrt{(\alpha+1)(\alpha+2)}}.$$
>
> "

Here $C_\alpha(p)$ is the norm of the inclusion from the Bloch space, with $\|f\|_{\mathcal{B}} = |f(0)| + \sup_{z\in\mathbb{D}}(1-|z|^2)|f'(z)|$, to A^p_α with probability measure $(\alpha+1)(1-|z|^2)^\alpha \,\mathrm{d}A(z)$, where $\mathrm{d}A = \mathrm{d}x\,\mathrm{d}y/\pi$. Source: Adrián Llinares, "Norms of Inclusions between Some Spaces of Analytic Functions," arXiv:2107.11315v2, Conjecture 3.9, §3, p. 9.

Answered: no. Evidence: Sketch.

No. For some $\eta < 1$ and $\varepsilon_0 > 0$, the limsup is at most $\eta/[2e\sqrt{(\alpha+1)(\alpha+2)}]$ whenever $-1 < \alpha < -1+\varepsilon_0$. The true limit and its existence remain undetermined.

Theorem 1. *There are constants $\eta < 1$ and $\varepsilon_0 > 0$ such that, for every $-1 < \alpha < -1+\varepsilon_0$,*

$$\limsup_{p\to\infty} \frac{C_\alpha(p)}{p} \le \frac{\eta}{2e\sqrt{(\alpha+1)(\alpha+2)}}. \tag{1}$$

Consequently the conjecture is false. This does not determine the actual limit or assert its existence.

The key is a uniform strict local average deficit of the Bloch quotient, as in Ivrii and Kayumov, *Makarov's principle for the Bloch unit ball*, equation (1.7) and §1.2. We prove the needed qualitative deficit and its weighted version below.

Put

$$\mathrm{d}\nu(z) = \frac{\mathrm{d}A(z)}{(1-|z|^2)^2}, \qquad q_f(z) = (1-|z|^2)|f'(z)|.$$

Let $\mathcal{U}$ consist of the analytic functions f on $\mathbb{D}$ with $f(0) = 0$ and $q_f \le 1$. Fix $\rho = 1/2$ and define

$$B_w = \left\{ z : \left| \frac{z-w}{1-\overline{w}z} \right| < \rho \right\}, \qquad V = \nu(B_w) = \frac{\rho^2}{1-\rho^2}, \qquad (\mathcal{A}v)(w) = \frac{1}{V}\int_{B_w} v(z)\,\mathrm{d}\nu(z).$$

The averaging operator $\mathcal{A}$ is positive and self-adjoint with respect to ν: its kernel is symmetric because $z \in B_w$ if and only if $w \in B_z$, and V does not depend on the center.

Lemma 1. *There is a constant $A < 1$ such that $\mathcal{A}(q_f^2) \le A$ for every $f \in \mathcal{U}$.*

Proof. The normalized family $\mathcal{U}$ is compact for locally uniform convergence: the derivative bound gives a uniform bound on every compact subset, and it is preserved under limits. In particular

$$A := \max_{f\in\mathcal{U}} \frac{1}{V}\int_{|z|<\rho} q_f(z)^2\,\mathrm{d}\nu(z)$$

exists. If $A = 1$, an attaining function would have $q_f = 1$ throughout that disk, since $q_f \le 1$ and is continuous. Its derivative would be nonvanishing and $\log|f'(z)| = -\log(1-|z|^2)$ there. The left side is harmonic and the right side has strictly positive Laplacian, a contradiction.

Date: 13 October 2026.

Thus $A < 1$; also $A > 0$, as is seen from $f(z) = z$. For an arbitrary center, compose f with a disk automorphism carrying 0 to that center and subtract its value at 0. The Bloch quotient and ν are invariant under that change of variables, proving the assertion. □

Lemma 2. *There are $B < 1$, $\varepsilon_1 > 0$, $0 < \tau_1 \le 1$, and $C < \infty$ such that, whenever $f \in \mathcal{U}$ is analytic on a neighborhood of $\overline{\mathbb{D}}$, $0 < \varepsilon \le \varepsilon_1$, and $0 \le \tau \le \tau_1$,*

$$\begin{aligned}\int_{\mathbb{D}} q_f(z)^2 e^{\tau \operatorname{Re} f(z)} (1-|z|^2)^{\varepsilon-1}\,\mathrm{d}A(z) \\ \le B\int_{\mathbb{D}} e^{\tau \operatorname{Re} f(z)} (1-|z|^2)^{\varepsilon-1}\,\mathrm{d}A(z) + C.\end{aligned} \tag{2}$$

All four constants are independent of f, ε, and τ in these ranges.

Proof. Set $L = \operatorname{atanh}\rho$. The Bloch derivative bound integrated along a hyperbolic geodesic gives $|f(z) - f(w)| \le L$ for $z \in B_w$. For

$$W(w) = e^{\tau \operatorname{Re} f(w)} (1-|w|^2)^{1+\varepsilon},$$

use the automorphism $z = (w+v)/(1+\overline{w}v)$ to obtain

$$\frac{\mathcal{A}W(w)}{W(w)} \ge e^{-\tau L} J_\varepsilon(w), \qquad J_\varepsilon(w) = \frac{1}{V}\int_{|v|<\rho} \left(\frac{1-|v|^2}{|1+\overline{w}v|^2}\right)^{1+\varepsilon} \mathrm{d}\nu(v). \tag{3}$$

The denominator in this integral is bounded away from zero, even for $|w| \le 1$. Hence $J_\varepsilon(w)$ is jointly continuous in $\varepsilon \ge 0$ and $w \in \overline{\mathbb{D}}$ near $\varepsilon = 0$. For $|w| = 1$, the angular average of $(1-|v|^2)/|1+\overline{w}v|^2$ is 1 on every circle $|v| = r$. It follows that $J_0(w) = 1$ on $|w| = 1$.

Choose $\kappa \in (A, 1)$, with A from Lemma 1. Joint continuity gives $r_0 < 1$ and $\varepsilon_1 > 0$ such that $J_\varepsilon(w) \ge \kappa$ when $r_0 < |w| < 1$ and $0 \le \varepsilon \le \varepsilon_1$. Choose $0 < \tau_1 \le 1$ small enough that

$$B \coloneqq \frac{A e^{\tau_1 L}}{\kappa} < 1.$$

By positivity, self-adjointness, and Lemma 1,

$$\int_{\mathbb{D}} q_f^2 \mathcal{A}W\,\mathrm{d}\nu = \int_{\mathbb{D}} W\mathcal{A}(q_f^2)\,\mathrm{d}\nu \le A\int_{\mathbb{D}} W\,\mathrm{d}\nu.$$

Combining this with (3) yields

$$\int_{|w|>r_0} q_f(w)^2 W(w)\,\mathrm{d}\nu(w) \le B\int_{\mathbb{D}} W\,\mathrm{d}\nu.$$

On the remaining disk, $q_f \le 1$ and $|f(w)| \le \operatorname{atanh} r_0$, so its contribution is bounded by

$$C = e^{\tau_1 \operatorname{atanh} r_0}\int_{|w|\le r_0} \frac{\mathrm{d}A(w)}{1-|w|^2} < \infty.$$

This is precisely (2). All integrals are finite for the stated functions, and the interchange of the nonnegative integrals is justified by Tonelli's theorem. □

Proof of Theorem 1. Choose D with $B < D < 1$, where B is as in Lemma 2. Take $0 < \varepsilon_0 \le \varepsilon_1$ small enough that

$$\tau = \tau(\varepsilon) \coloneqq \frac{2\sqrt{\varepsilon(1+\varepsilon)}}{\sqrt{D}} \le \tau_1 \qquad (0 < \varepsilon < \varepsilon_0). \tag{4}$$

Fix one such ε. We first prove

$$\sup_{f\in\mathcal{U}} \int_{\mathbb{D}} e^{\tau \operatorname{Re} f(z)} (1-|z|^2)^{\varepsilon-1}\,\mathrm{d}A(z) < \infty. \tag{5}$$

Initially assume f is analytic past $\overline{\mathbb{D}}$, and write

$$F = e^{\tau \operatorname{Re} f}, \qquad M = \int_{\mathbb{D}} F(1-|z|^2)^{\varepsilon-1}\,\mathrm{d}A, \qquad N = \int_{\mathbb{D}} F(1-|z|^2)^{\varepsilon}\,\mathrm{d}A,$$

and let E denote the left side of (2). Using the Euclidean Laplacian,

$$\Delta F = \tau^2 |f'|^2 F, \qquad \Delta(1-|z|^2)^{\varepsilon+1} = 4\varepsilon(\varepsilon+1)(1-|z|^2)^{\varepsilon-1} - 4(\varepsilon+1)^2(1-|z|^2)^{\varepsilon}.$$

Green's identity therefore gives

$$\tau^2 E = 4\varepsilon(\varepsilon+1)M - 4(\varepsilon+1)^2 N. \tag{6}$$

One may apply the identity on $|z| < r$ and let $r \uparrow 1$: F and its first derivatives are bounded, while the weight and its normal derivative are $O((1-r^2)^{\varepsilon+1})$ and $O((1-r^2)^{\varepsilon})$, respectively. Thus both boundary terms vanish; the interior integrals converge because $\varepsilon > 0$.

The radial Bloch bound gives

$$|f(z)| \le \operatorname{atanh}|z| \le \log 2 - \frac{1}{2}\log(1-|z|^2).$$

Consequently, since $\tau \le 1$,

$$N \le \frac{2^\tau}{1+\varepsilon-\tau/2} \le 4.$$

Combining (2) and (6) now gives

$$\bigl(4\varepsilon(\varepsilon+1) - B\tau^2\bigr)M \le \tau^2 C + 16(\varepsilon+1)^2.$$

The coefficient on the left is $4\varepsilon(\varepsilon+1)(1-B/D) > 0$ by (4). This bounds M uniformly over all the stated f. For arbitrary $f \in \mathcal{U}$, its dilates $f(rz)$ are analytic past $\overline{\mathbb{D}}$ and remain in $\mathcal{U}$, since

$$(1-|z|^2)r|f'(rz)| \le \frac{r(1-|z|^2)}{1-r^2|z|^2} \le 1.$$

Fatou's lemma proves (5). It also holds, with the same constant, for every rotation $e^{i\theta}f$.

For any $0 < s < \tau$, choose finitely many angles θ_j such that $\max_j \operatorname{Re}(e^{i\theta_j} w) \ge (s/\tau)|w|$ for every $w \in \mathbb{C}$. Then

$$e^{s|f(z)|} \le \sum_j e^{\tau \operatorname{Re}(e^{i\theta_j} f(z))}.$$

Thus (5) implies uniform exponential integrability of $e^{s|f|}$ over $\mathcal{U}$ against the probability measure $\varepsilon(1-|z|^2)^{\varepsilon-1}\,\mathrm{d}A$. If $\|f\|_{\mathcal{B}} \le 1$, then $f - f(0) \in \mathcal{U}$ and $|f(0)| \le 1$, so the same assertion holds over the full Bloch unit ball. Since $x^p \le (p/(es))^p e^{sx}$ for $x \ge 0$, there is a finite constant $K_{\varepsilon,s}$, independent of p and f, such that

$$C_{\varepsilon-1}(p) \le \frac{p}{es} K_{\varepsilon,s}^{1/p}.$$

First let $p \to \infty$, and then $s \uparrow \tau$. We obtain

$$\limsup_{p\to\infty} \frac{C_{\varepsilon-1}(p)}{p} \le \frac{1}{e\tau} = \frac{\sqrt{D}}{2e\sqrt{\varepsilon(1+\varepsilon)}}.$$

Taking $\eta = \sqrt{D} < 1$ and $\alpha = \varepsilon - 1$ proves (1). □